INDIAN STATISTICAL INSTITUTE, KOLKATA
BACHELOR OF STATISTICS

Psychology

Semester Project

Analysis Of Current Youth Mindsets Towards Nuclear Weapons

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1 Aim of the Project

The aim of this project is to understand the attitudes of college students toward nuclear weapons and contemporary geopolitical issues such as war, and to explore factors associated with these attitudes..

2 Description of Dataset

The data were collected from college students belonging to various colleges across Andhra Pradesh. The dataset contains demographic variables such as gender, family income, mother tongue, religion, etc., along with responses to 50 Likert-scale questions related to nuclear weapons and geopolitical issues.

Each Likert item had five response categories:
Strongly Agree, Agree, Neither Agree nor Disagree, Disagree, and Strongly Disagree.

A total of 35 responses were initially collected.
The questions can be accessed via this link: Questionnaire

3 Data Cleaning

To identify inconsistent or careless responses, some trap questions were given in the questionnaire, ie. similar questions or their negations were put to check the correctness of the data. And after cleaning the data, we were left with 33 responses.

4 Descriptive Statistics

The following observations were made

1. From the data we have observed that 60% people either disagree or strongly disagree with the idea that possessing nuclear weapons is a burden on Indians, given India's Economic Conditions.

2. 96% respondents agree or strongly agree that India's nuclear weapons program has prevented others from attacking India.

3. 72% of respondents agree or strongly agree that there should be international controls on more countries having nuclear weapons.

4. Around only 18% respondents disagree that Nuclear-armed countries should lower the number of nuclear weapons they have.

5. Around 93% people believe that there should be more efforts on educating people about the effects of nuclear weapons.

6. Around 42% respondents agree or strongly agree that if all other countries, including U.S.A. and Russia, give up their nuclear weapons, India should also give up its nuclear weapons. and around 24% of respondents disagree with it.

7. Around 73% people Disagree or Strongly Disagree that if Pakistan and China give up their nuclear weapons India should also give up its nuclear weapons and 12% Agrees with it.

8. 60% of respondents Disagrees or Strongly Disagrees with the statement "Nuclear weapons cannot destroy the world".

9. Also, we observed that 42% respondents have agreed or strongly agreed that they would immigrate to a new country if they had a better career opportunity and 21% respondents have Disagreed or Strongly Disagreed.

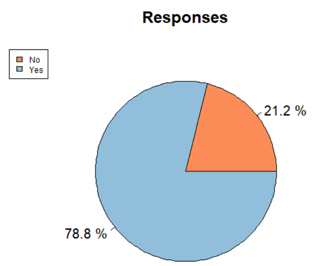
10. Roughly 79% people agree or strongly agree that India should start compulsory military service at the age of 18 for 1 year.

11. 48% respondents agree or strongly agree that India should focus more on developing conventional military capabilities rather than investing in nuclear and around 24% disagree or strongly disagree.

12. And in the response to the statement "Pakistan having nuclear weapons has made Pakistanis safer." around 30% respondents agree or strongly agree, 30% respondents disagree or strongly disagree and around 40% respondents neither agree nor disagree.

13. 60% respondents agree or strongly agree that Nuclear weapons play a role in maintaining global peace.

14. In response to the question "Would your answers be the same, if you belonged to a non-nuclear country?", 78% respondents said "Yes".



5 Testing Association Among Questions Asked

5.1 Statistical Method Used

Kendall's Tau Test has been used to test association among two variables(here questions), because Kendall's Tau Test works well for small datasets and it's useful for analyzing ordinal data like Likert scale, etc.
Kendall's Tau test is a non-parametric test procedure. In R Kendall's Tau is calculated as:

$$\tau = \frac{C - D}{\sqrt{(C + D + T_x)(C + D + T_y)}}$$

where C is the number of concordant pairs and D is the number of discordant pairs.And T_x is Ties in variable X, T_y is Ties in variable Y
In our hypothesis testing
H_o : The correlation coefficient $\tau = 0$ (there is no correlation).
H_1 : the correlation coefficient$\tau \neq 0$ (there is a correlation).
In R, a test statistic Z based on Kendall's Tau is used for computing asymptotic p-values, such that under the null hypothesis, the statistic asymptotically converges in distribution to $N(0, 1)$. Its explicit form is quite complicated.

For the present analysis, asymptotic p-values were used as an approximation because the Likert-scale data contain ties. Reject H_o if asymptotic $p - value < 0.05$.
When multiple hypothesis tests are performed, the probability of obtaining false positive results increases, leading to an increased overall Type I error rate.

The **Benjamini-Hochberg** procedure addresses this issue by controlling the FDR, which is defined as the expected proportion of false positives among the rejected hypotheses.
The calculation of adjusted p-values in the Benjamini-Hochberg procedure involves comparing each individual p-value to a critical value or threshold.
Step 1: Collect all $p - values$ from your tests.
Step 2: Sort the p-values in ascending order.
Step 3: Assign ranks (1 for smallest $p - value$, up to n).
Step 4: Choose your FDR level (e.g., $Q = 0.05$).
Step 5: For each p-value, calculate:

$$Threshold = \frac{Rank \times Q}{n}$$

Step 6: Find the largest$p - value \leq$ its $Threshold$.
Step 7: Reject all hypotheses with $p - values \leq$ that value
The adjusted p-values provide a corrected measure of statistical significance that accounts

for multiple testing.
Note that,Benjamini-Hochberg procedure assumes that the null hypotheses are independent or exhibit positive dependency.
(Thus here instead of using $p-values$ we have used adjusted $p-values$, for rejecting H_o)

5.2 Observations

The following questions/statements were found to be significantly associated ($p < 0.05$) with the statement "*Given the economic status of Indians, possessing nuclear weapons is a burden on Indians.*"(call it ST17)

1. "Indian States should allow mass immigration", $\tau \approx 0.43$ and $p-value \approx 0.0042 < 0.05$. Suggests a positive association with ST17.

2. "India should get rid of its nuclear weapons.", $\tau \approx 0.76$ and $p-value \approx 1.065e-06$. Suggests a positive association with ST17.

3. "If Pakistan gives up their nuclear weapons India should also give up its nuclear weapons.", $\tau \approx 0.51$ and $p-value \approx 0.0008$. Suggests a positive association with ST17.

4. "India should welcome permanent migrants from other countries.", $\tau \approx 0.44$ and $p-value \approx 0.003$. Suggests a positive association with ST17

5. "If Pakistan and China give up their nuclear weapons India should also give up its nuclear weapons.", $\tau \approx 0.44$ and $p-value \approx 0.004$. Suggests a positive association with ST17.

6. "India's nuclear weapons program has prevented others from attacking India.", $\tau \approx -0.49$ and $p-value \approx 0.002 < 0.05$. Suggests a negative association with ST17.

7. "If Pakistan and China give up their nuclear weapons India should also give up its nuclear weapons.",$\tau \approx -0.46$ and $p-value \approx 0.002 < 0.05$. Suggests a negative association with ST17.

The following questions/statements were found to be significantly associated ($p < 0.05$) with the statement "*If Pakistan gives up their nuclear weapons India should also give up its nuclear weapons.*"(call it ST24)

1. "India's nuclear weapons program has prevented others from attacking India.",$\tau \approx -0.477$ and $p-value \approx 0.003 < 0.05$. Suggests a negative association with ST24.

2. "If Pakistan and China give up their nuclear weapons, India should also give up its nuclear weapons.",$\tau \approx 0.63$ and $p-value \approx 3.641e-05 < 0.05$. Suggests a positive association with ST24.

3. "I will immigrate to a foreign country if I have better career opportunities.",$\tau \approx -0.44$ and $p-value \approx 0.003 < 0.05$. Suggests a negative association with ST24.

The following questions/statements were found to be significantly associated ($p < 0.05$) with the statement "*A nuclear weapon will cause killing of 1,00,000 people or more.*"(call it ST29)

1. "Possibility of accidents in a nuclear plant is a significant concern.",$\tau \approx 0.51$ and $p-value \approx 0.002 < 0.05$. Suggests a positive association with ST29.

2. "The use of nuclear weapons can lead to the end of the world.",$\tau \approx 0.51$ and $p-value \approx 0.001 < 0.05$. Suggests a positive association with ST29.

3. "India should focus more on using nuclear technology for generating electricity, rather than weapons.",$\tau \approx 0.52$ and $p-value \approx 0.0009 < 0.05$. Suggests a positive association with ST29.

The following questions/statements were found to be significantly associated ($p < 0.05$) with the statement "*India should welcome permanent migrants from other countries.*"(call it ST40)

1. "Given the economic status of Indians, possessing nuclear weapons is a burden on Indians."(ST17)

2. "Indian states should allow mass immigration from other states.",$\tau \approx 0.62$ and $p-value \approx 2.733e-05 < 0.05$. Suggests a positive association with ST40.

3. "Current Indian media is neutral about politics in India.",$\tau \approx 0.47$ and $p-value \approx 0.001 < 0.05$. Suggests a positive association with ST40.

The following questions/statements were found to be significantly associated ($p < 0.05$) with the statement "*India should focus more on developing conventional military capabilities rather than investing in nuclear*"(call it ST54)

1. "India should be more transparent about its nuclear weapons program to the Indian public.",$\tau \approx 0.44$ and $p-value \approx 0.002 < 0.05$. Suggests a positive association with ST54.

2. "Current constitution of India should be radically changed.",$\tau \approx 0.51$ and $p-value \approx 0.0004 < 0.05$. Suggests a positive association with ST54.

The following questions/statements were found to be significantly associated ($p < 0.05$) with the statement "*A nuclear weapon will cause killing of 1,00,000 people or more.*"(call it ST60)

1. "A nuclear weapon will cause the killing of 1,00,000 people or more."(ST29)(Positive association)

2. "The use of nuclear weapons can lead to the end of the world.",$\tau \approx 0.60$ and $p - value \approx 6.929e - 05 < 0.05$. Suggests a positive association with ST60.

3. "There should be more efforts to educate the public about the effects of nuclear weapons.",$\tau \approx 0.44$ and $p - value \approx 0.005 < 0.05$. Suggests a positive association with ST60.

4. "India should be more transparent about its nuclear weapons program to the Indian public.",$\tau \approx 0.50$ and $p - value \approx 0.0007 < 0.05$. Suggests a positive association with ST60.

5. "Current constitution of India should be radically changed.",$\tau \approx 0.46$ and $p - value \approx 0.002 < 0.05$. Suggests a positive association with ST60.

Next, the **Mann-Whitney U test (Wilcoxon rank-sum test)** was used to examine whether responses to the questions differed significantly across gender groups. This test was used because gender is a categorical nominal variable, whereas the questionnaire responses are measured on an ordinal Likert scale. The following questions were found to show a statistically significant association with gender:

1. "Given the economic status of Indians, possessing nuclear weapons is a burden on Indians."

2. "India's nuclear weapons program has prevented others from attacking India."

3. "India should continue to possess nuclear weapons."

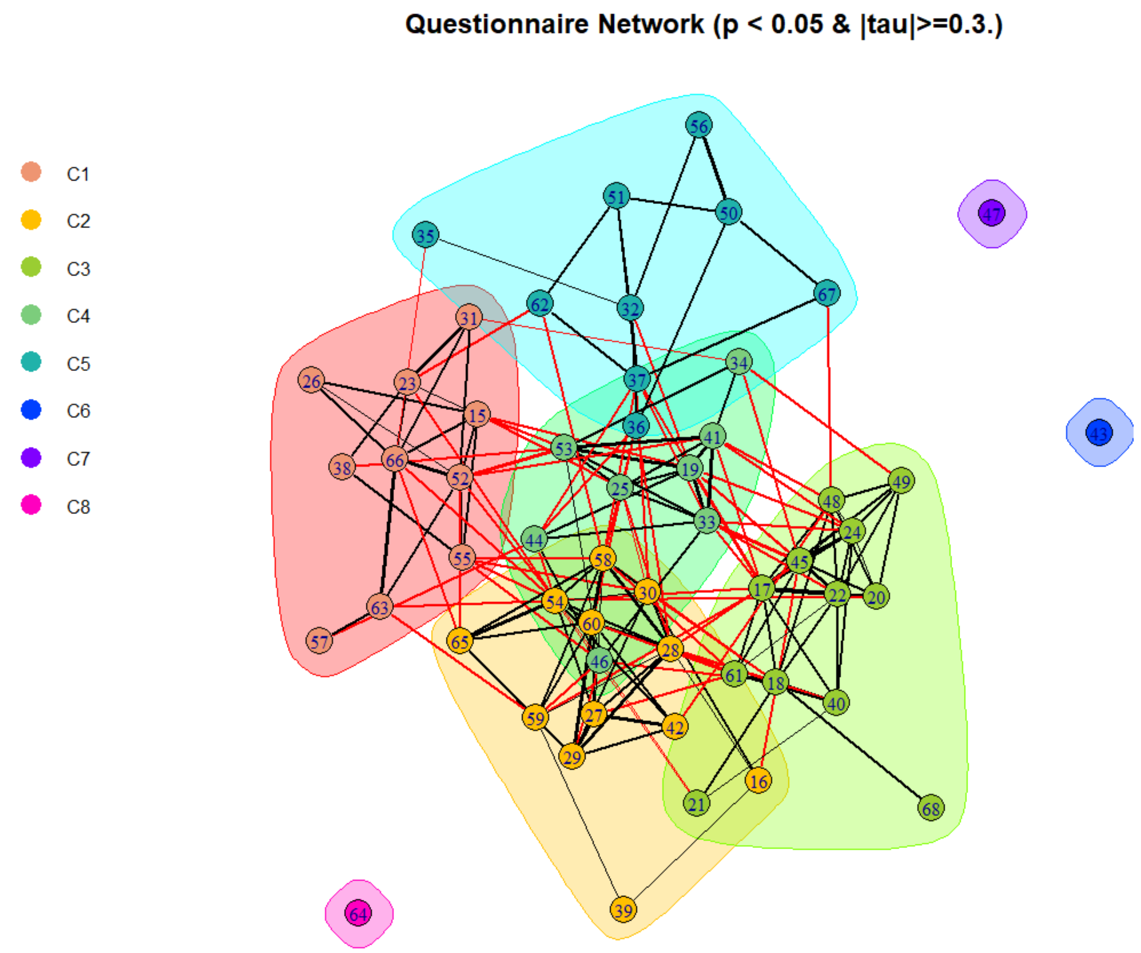
4. "Nuclear weapons that can kill many more people instantly should be developed."

5. "Couples should stay with boy's parents after marriage."

6. "I will immigrate to a foreign country if I had better career opportunities."

6 Questionnaire Network

A network was plotted where the vertices are the likert questions and the V_1 and V_2 have a edge between them iff p-value of kendall's tau test between them is < 0.05 and $|\tau| > 0.3$.



The numbers are the the serial numbers of the questions in the *original questionnaire.* The red colour of the edge represents negative associations between vertices(questions), and black represents positive association. The thickness of the edge is directly proportional to the strength of association.

Lovain Method for community detection is used to form clusters.The Louvain method works by repeating two phases. In phase one, nodes are sorted into communities based on how the modularity of the graph changes when a node moves communities. In phase two, the graph is reinterpreted so that communities are seen as individual nodes.we consider how moving v from its current community C into a neighboring community C' will affect the modularity of the graph partition.We select the community C' with the greatest change in modularity, and if the change is positive, we move v into C'; otherwise we leave it where it is. This continues until the modularity stops improving. Here modularity is :

$$Q = \frac{1}{2m}\sum_{i=1}^{N}\sum_{j=1}^{N}[A_{ij} - \frac{k_i k_j}{2m}]\delta(c_i, c_j)$$

where:

- A_{ij} represents the edge weight between nodes i and j;

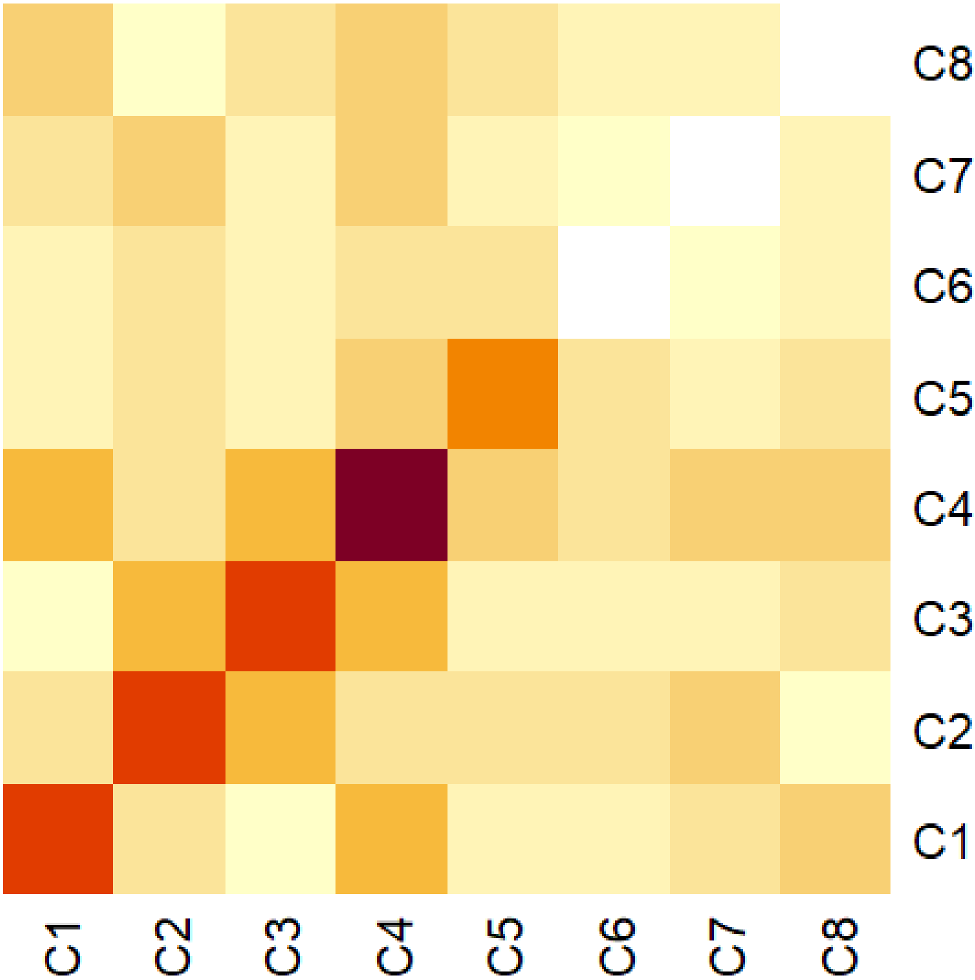
- k_i and k_j are the sums of the weights of the edges attached to nodes i and j, respectively;
- m is the sum of all edge weights in the graph;
- N is the total number of nodes in the graph;
- c_i and c_j are the communities to which nodes i and j belong; and
- δ denotes the Kronecker delta function.

6.1 Interpretations

After froming communities it has been observed that :

1. Community 1(C1) represents Psychological & Spiritual Traits
2. Community 2(C2) represents Nuclear Fear, Risk & National Security Skepticism
3. Community 3(C3) represents Anti-Nuclear & Internationalist Orientation.
4. Community 4(C4) represents Pro-Nuclear National Security Attitudes.
5. Community 5(C5) represents Institutional Trust & Civic Attitudes.
6. Community 6(C6) represents Centralized Nuclear Decision-Making.
7. Community 7(C7) represents No First Use Doctrine.
8. Community 8(C8) represents Nuclear Weapons Maintain Peace

Below is the Association between communities Heat Map.



Darker colour represents stronger association, Lighter colour represents weaker association.

7 Conclusion

The present study explored the attitudes of college students toward nuclear weapons, national security, migration, governance, and related geopolitical issues using questionnaire-based survey data. Descriptive analysis suggested that a large proportion of respondents viewed nuclear weapons as important for India's security, while many also supported international regulation, public awareness, and transparency regarding nuclear weapons. Significant associations were observed among several ideological, political, and social attitudes using Kendall's Tau based association analysis.

The network analysis further revealed meaningful community structures among questionnaire items. Distinct clusters related to psychological and spiritual traits, nuclear risk perception, anti-nuclear internationalist attitudes, pro-nuclear national security attitudes, and institutional or civic beliefs were identified using the Louvain community detection method. These findings suggest that opinions regarding nuclear weapons are connected not only with views on national security, but also with broader social, political, and personal belief systems.

Since the study was conducted on a relatively small sample, the results should be interpreted cautiously and viewed as exploratory rather than definitive. Nevertheless, the analysis demonstrates how statistical and network-based methods can be used to study complex patterns of attitudes and ideological structures within survey data. Future studies with larger and more diverse samples may help validate and extend these findings.

R-code link: Psychology-Project-Questionaire./blob/main/R-code.R

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